Mathematical pendulum in the field of a ball

X16 (2016) — English translation

A mathematical pendulum makes small oscillations near the surface of a dielectric ball uniformly charged with negative electricity. The length of the pendulum is equal to the radius of the ball.

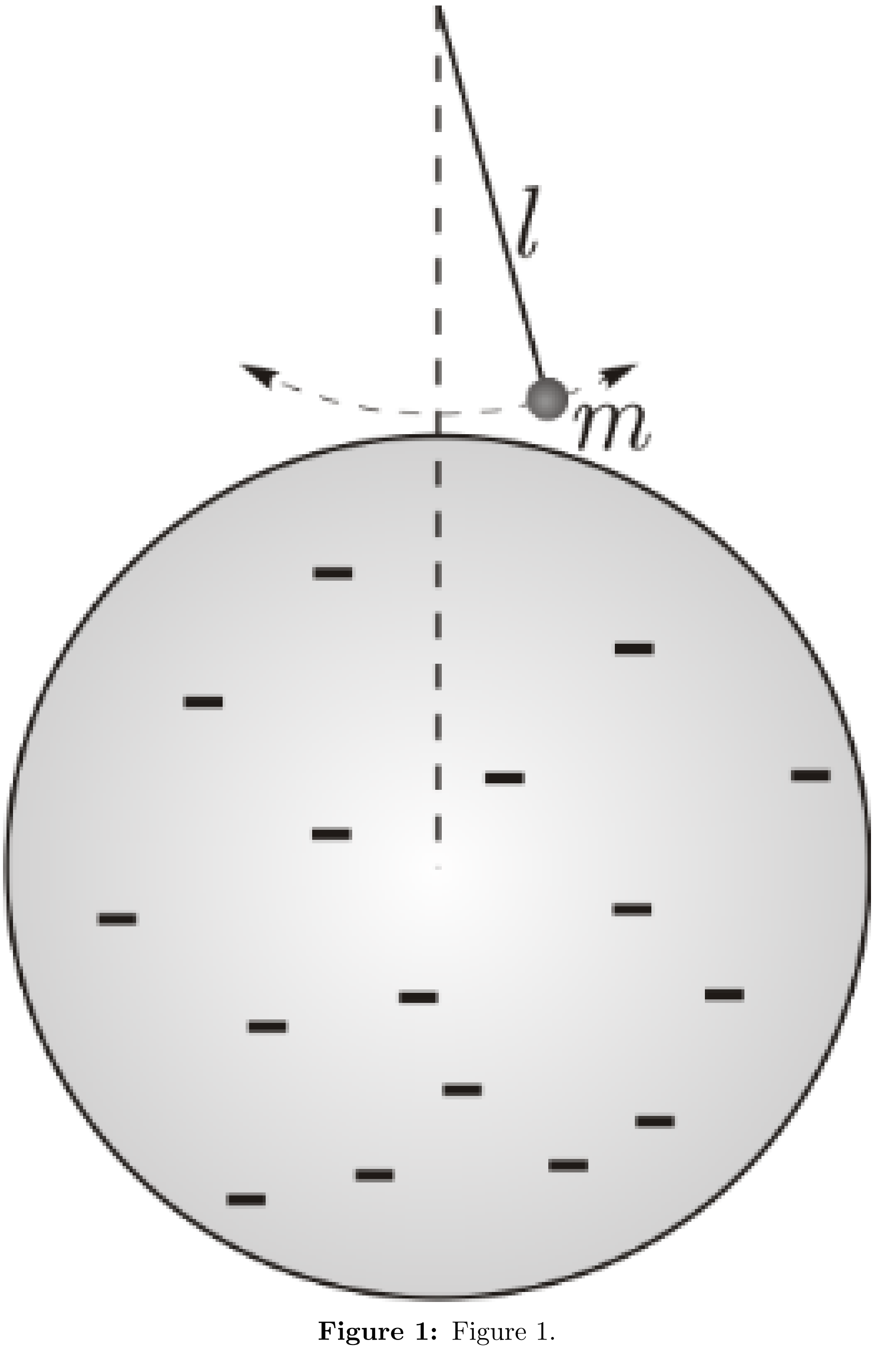

Figure 1: Figure 1.

A1	What is the period of harmonic oscillations of a pendulum if the length of its thread is l, the mass of the weight is m, the charge is $q > 0$, and the magnitude of the field strength near the surface of the ball is E_0.	**2.00**

Ignore gravity and the field of induction charges

Source: `https://pho.rs/p/3998`